

Diophantine Equations

Problem 1. Prove that if p and q are two prime numbers such that

$$p + p^2 + p^3 + \cdots + p^q = q + q^2 + q^3 + \cdots + q^p,$$

then $p = q$.

Problem 2. For any integer $n > 3$ denote by $n?$ the product of all primes less than n . Solve the equation $n? = 2n + 16$.

Problem 3. Solve the Diophantine equation $x^{2018} - y^{2018} = (xy)^{2017}$ where x and y are nonnegative integers.

Problem 4. Show that no pairs of integers (m, n) satisfy $2560m^2 + 5m + 6 = n^5$.

Problem 5. Find all positive integers x and y such that $x^{x+y} = y^{3x}$.

Problem 6. Find all pairs (p, q) of prime numbers satisfying

$$p^3 + 7q = q^9 + 5p^2 + 18p.$$

Problem 7. Find all pairs of integers (m, n) which satisfy the equation

$$(2n^2 + 5m - 5n - mn)^2 = m^3n.$$

Problem 8. Find all positive integers a, b, c and prime p satisfying that

$$2^a p^b = (p + 2)^c + 1.$$